



Experimental and numerical investigation of single-phase flow and heat transfer characteristics in counter-flow manifold microchannel heat sink

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ABSTRACT

The miniaturization of high-power electronics results in significantly increased heat flux, making thermal management critical. Manifold microchannel heat sinks are a key research focus due to their compact design and superior heat transfer capability. However, they typically exhibit large temperature gradients and high pressure drops. To mitigate these issues, a counter-flow manifold microchannel heat sink (CFMMC) is designed. This configuration incorporates a flow distribution plate to enable secondary fluid redistribution, thereby establishing localized counter-flow within the microchannels. To evaluate the flow and thermal performance of the CFMMC, this study employs a flow visualization experimental platform. Deionized water is used as the working fluid. Tests are conducted at total heat fluxes of 50, 60, and 70 W/cm² with a constant inlet temperature of 298.15 K. The flow and heat transfer characteristics of the CFMMC are compared with the parallel-flow manifold microchannel heat sink (PFMMC) over a mass flow rate range of 80 g/min to 170 g/min. The results show a more uniform flow distribution in the CFMMC compared to the PFMMC. The mass flux within individual channels is lower and fluctuates less. The maximum mass flux within the CFMMC channels is approximately 61% lower than that in the PFMMC channels. The CFMMC exhibits a 23.11% reduction in the maximum maldistribution index. Under all experimental conditions, the CFMMC exhibits a lower overall pressure drop and friction factor. The pressure drop is approximately 23% lower than that of the PFMMC. The CFMMC also provides enhanced cooling performance with a smaller average wall temperature gradient. It yields higher heat transfer coefficients and Nusselt numbers. At a heat flux of 50 W/cm² and a mass flow rate of 170 g/min, the average bottom wall temperature of the CFMMC channels is approximately 4 K lower compared to the PFMMC. The thermal resistance is reduced by about 15%. The heat transfer coefficient is improved by 24.4%. Under identical heat flux conditions, the CFMMC achieves a higher coefficient of performance than the PFMMC. Across the three tested heat flux levels, the performance evaluation criterion (PEC) for the PFMMC remains below 1.5 and approaches 1 under some conditions. In contrast, the PEC for the CFMMC mostly exceeds 1.4 and remains more stable at higher flow rates. This demonstrates its superior overall thermo-hydraulic performance. In the extreme performance test, under single-phase conditions and at a mass flow rate of 2000 g/min, the PFMMC can withstand a maximum effective heat flux of 175 W/cm². The CFMMC withstands 300 W/cm² while achieving an approximate 54% reduction in pressure drop compared to the PFMMC.

1. Introduction

With the rapid advancement of electronics in the information age, the continuous miniaturization of high-power devices has led to a sharp increase in heat flux per unit area [1,2]. This makes thermal dissipation a primary cause of failure in high-power electronic components [3]. Therefore, developing efficient and reliable thermal management technologies is critical for ensuring the stable operation of high-power

electronics [4–6]. Microchannel heat sinks offer several distinct advantages over other cooling methods. These include a high heat transfer area per unit fluid flow, compact dimensions, low coolant volume requirements, and a relatively uniform wall temperature distribution [7]. Consequently, performance optimization of microchannel heat sinks has become a key research topic in thermal management.

In 1981, Tuckerman and Pease [8] first proposed the microchannel heat sink. They demonstrated heat dissipation of up to 790 W/cm² using

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water as the coolant in straight parallel microchannels. Straight parallel microchannels often suffer from non-uniform flow distribution, temperature variation, and high pressure drop. Therefore, Harpole and Eninger [9] first proposed the concept of a manifold microchannel heat sink (MMC) in 1991. The research found that the manifold arrangement is used to distribute flow through alternating inlet and outlet pairs, which significantly reduces the pressure drop inherent in conventional fluid delivery systems. It also enhances the heat transfer coefficient (HTC) by restricting the growth of the thermal boundary layer. van Erp et al. [10] identified significant flow maldistribution within manifold microchannels. They effectively reduced this inter-channel flow imbalance by modifying the cross-sectional area of the channels. Many researchers have optimized microchannel performance through various approaches. These include modifying the manifold structure [11,12], altering the microchannel structure [13–17], incorporating fins and cavities [18–20], applying surface treatments [21–23], changing the coolant [24,25].

For most single-pass microchannels, particularly those with relatively long flow paths, a significant rise in wall temperature along the channel is commonly observed. Under two-phase flow boiling conditions, the increase in vapor quality along the microchannel, coupled with rapid bubble expansion, restricts sufficient liquid supply to downstream regions. Prolonged dryout in the outlet region can trigger premature critical heat flux (CHF), thereby degrading heat transfer performance [26]. Many researchers have achieved an improvement in the performance of microchannel heat sinks by establishing counter-current flow. Vafai and Zhu [27] first introduced a novel counter-flow double-layer microchannel heat sink. Their results demonstrated that, compared to a single-layer design, the counter-flow configuration significantly reduces both thermal resistance and fluid temperature rise. It also achieves a markedly lower pressure drop. Beh et al. [28] found that the temperature distribution in a double-layer microchannel differs from that in a single-layer one. The highest temperature occurs at the mid-region of the bottom wall in the double-layer configuration. In contrast, the peak temperature appears near the outlet of the bottom wall in the single-layer microchannel. Andhare et al. [29] found that the performance of a multi-pass manifold microchannel-based heat exchanger in a counter-flow configuration is enhanced in low heat flux applications. At flow rates as low as 20 g/s, the overall heat transfer coefficient reaches approximately 20,000 W/m²·K. This corresponds to a single-side heat transfer coefficient of about 60,000 W/m²·K for the heat exchanger. Shen et al. [30] proposed a modification of double-layer microchannel heat sinks (DMHSS) into wavy configuration with swap of upper and lower layer design. They demonstrated that this design effectively improves the comparatively low cooling effectiveness at the upper layer, and that the flow swap promotes the re-development of the flow field and enhances heat transfer accordingly. Radwan et al. [31] found that the microchannel heat sink design with parallel flow configuration attained effective cooling only at higher flowrates compared to the counter flow configuration; the counter flow configuration was superior at lower flow rates. Jung et al. [32] integrated an embedded silicon microchannel cold plate with a 3D manifold. Their study demonstrated that a heat flux of 300 W/cm² can be dissipated under single-phase conditions using the working fluid R-245fa. In addition, they also found that it provided more temperature uniformity across large area chips and it was less prone to flow instability in two-phase boiling heat transfer. Patel et al. [33] proposed a double-layer microchannel and conducted single-phase flow experiments. Their results showed that the counter-flow microchannel configuration eliminates downstream wall overheating. Li et al. [34] proposed a bidirectional counter-flow microchannel heat sink (BCF). Compared with single-phase conventional unidirectional parallel-flow microchannel heat sink (UPF), the BCF demonstrated significant advantages. It effectively avoids excessive downstream temperature rise by supplying liquid from both directions. This mechanism delays the premature formation of CHF. In their experiments, the CHF and average heat

transfer coefficient were enhanced by 33.8 %–57.2 % and 36.6 %–56.7 %, respectively. Wang et al. [35] proposed a microchannel heat sink with counter-flow interconnection configuration (CFIM). The study found that the interconnected slots induced fluid mixing between adjoining channels, which counteracted the low vapor quality in upstream channels with high vapor quality in downstream channels, thus notably ameliorating the average heat transfer coefficient performance. In their study, a significant enhancement in heat transfer performance was achieved in CFIM compared with co-current microchannel (CCM) and counter-flow microchannel (CFM). The average two-phase HTC was increased by 70.4 %–83.6 % at 40 °C, and the mass flux ranged from 291 to 618 kg/m²·s. Pei et al. [36] proposed a counter-flow microchannel with sinusoidal wavy walls (CFSWM). They reported that the counter-flow design enhanced thermal interaction between adjacent channels, delayed heat transfer deterioration caused by downstream wall overheating and eliminated backflow under all operating conditions. In their study, the pressure drop was reduced by 21 %, while the coefficient of performance (COP) increased by 36 %. Additionally, wall temperature uniformity was significantly improved, with the maximum wall temperature difference decreasing by 74 %. Xia et al. [37] performed a multi-objective optimization on a diamond-based U-type counter-flow manifold microchannel heat sink. Their study found that the fin width exerted the greatest influence on thermal resistance, accounting for 71.49 % of the total variance. The microchannel width was the primary factor affecting total entropy generation and pumping power. The multi-objective optimization scheme achieved an optimal trade-off between thermal resistance and pumping power. It reduced thermal resistance by 44.1 % while limiting the increase in pumping power to 13.3 %. Jiang et al. [26] proposed a counter-flow diverging manifold microchannel heat sink. A counter-flow configuration facilitates the channel-to-channel heat transfer to enhance bubble nucleation from the gradient cavities in the entrance region with subcooled liquid, while suppressing void growth near the channel exit.

To achieve the counterflow effect, traditional integrated counterflow manifold microchannels typically have complex manifold structures, making them difficult to manufacture and resulting in significant pressure drops. While simple multi-inlet manifold designs can improve fluid flow distribution, they do not change the flow direction within the channel and have limited impact on improving temperature uniformity. Achieving better temperature uniformity, better cooling performance, and lower pump power remains a major challenge for heat sinks of high power density electronic devices. Therefore, we propose integrating the localized counter-flow pattern with a multi-inlet manifold configuration. A four-inlet and four-outlet manifold configuration is adopted. A flow distribution plate with staggered distribution holes is introduced between the manifold and the microchannels. In summary, the fluids in adjacent channels flow in opposite directions, exhibiting counter-current flow, which can greatly improve the temperature uniformity of the manifold microchannels. Secondly, before entering the microchannels, the fluid must pass through the staggered distribution holes on the flow distribution plate for a second flow distribution, so that the fluid enters each channel more evenly, thereby improving the uniformity of flow distribution from the source. In previous numerical studies [38,39], the introduction of a counter-flow region was found to improve flow distribution in Z-type manifold layouts. This enhancement boosts the overall thermal performance of the manifold microchannel heat sink. Specifically, the maximum temperature can be reduced by approximately 5 K. Additionally, the pressure drop in the Z-type counter-flow manifold microchannel heat sink is 27.13 % lower than that in the conventional Z-type manifold design. Based on previous studies, we construct an experimental setup for a counter-flow manifold microchannel heat sink (CFMMC). A comparison is made with the parallel-flow manifold microchannel heat sink (PFMMC). The comprehensive performance of both configurations is evaluated under varying flow rates and heat fluxes.

2. Experimental system and methodology

2.1. Flow loop system

Fig. 1 illustrates the schematic of the visualized MMC heat transfer experimental setup. Fig. 2 provides a photograph of the actual system. The system primarily consists of the flow loop, the heating system, the visualization system, and the data acquisition system. Deionized water is used as the working fluid, with its inlet temperature maintained at 298.15 K. During the experiment, the working fluid flows from the thermostatic water bath (temperature control accuracy ± 0.5 K) and passes through a filter. A gear pump then drives the fluid through a mass flow meter into the test section. Heating in the test section is provided by nine electric heating rods connected to a DC power supply. Temperature and pressure drop during the experiment are measured by thermocouples and a differential pressure transducer, respectively. The working fluid subsequently passes through a heat exchanger, where it exchanges heat with a low-temperature medium from the other thermostatic water bath before returning to the bath, completing the closed loop. The measured flow rate, temperature, and pressure drop data are collected by a data acquisition instrument and transmitted to a computer for monitoring. The main equipment and instruments are listed in Table 1.

2.2. Test section

Fig. 3 presents the schematic diagram of the experimental test section. The PFMMC consists of four parts: an upper clamping plate, a lower clamping plate, a monolithic copper microchannel substrate and the manifold. The monolithic copper microchannel is fabricated from a high-thermal-conductivity copper ($k = 387.6$ W/(m·K)) block using CNC machining. Its upper section forms the microchannel region. This region contains 20 parallel channels. Each channel is 20 mm in length, 0.5 mm in width, and 1 mm in depth. The spacing between adjacent channels is 0.5 mm. The lower section serves as the heating block. It incorporates

nine cylindrical grooves. Each groove has a diameter of 8 mm and a depth of 50 mm. These grooves house heating rods to provide the heat input for the experimental system. Each heating rod has a rated power of 230 W. Three thermocouple holes are located beneath the channels. Each hole has a diameter of 1.5 mm and a depth of 20 mm. The center of each hole is positioned 1.75 mm from the bottom surface of the channel. The distance between adjacent holes is 8.25 mm. These three thermocouple holes are equally spaced along the centerline in the flow direction (corresponding to the inlet, middle, and outlet sections, respectively). They are arranged to acquire the average temperature of the test section for the calculation of the average thermal performance and thermal resistance. The manifold is made of polysulfone, chosen for its high-temperature resistance and good optical transparency. The primary purpose of employing the low-thermal-conductivity polysulfone manifold is to enable visual observation. Concurrently, from a thermal perspective, it acts as a top insulating layer, effectively reducing heat loss. The manifold channels are fabricated via CNC machining. Each manifold channel has a width of 2 mm and a depth of 1 mm. A circular sealing groove is also machined on the manifold. This sealing groove accommodates a silicone gasket to prevent working fluid leakage. The upper and lower clamping plates are manufactured from stainless steel. Each plate contains twenty bolt holes with a diameter of 4 mm. The monolithic copper microchannel substrate and the manifold are clamped together by these plates. The assembly is secured using bolts. The manifold channel dimensions in the CFMMC are identical to those in the PFMMC. The width and depth remain 2 mm and 1 mm, respectively. However, the PFMMC uses a two-inlet and two-outlet manifold configuration. In contrast, the CFMMC uses a four-inlet and four-outlet manifold configuration. Furthermore, a thin flow distribution plate with a thickness of 0.4 mm is inserted between the microchannel and the manifold. This plate incorporates 80 staggered holes, each measuring 2 mm in length and 0.5 mm in width. The misalignment spacing between the diversion holes is 0.5 mm. The lateral spacing between holes in the same row is 1.5 mm, while the longitudinal spacing between holes along

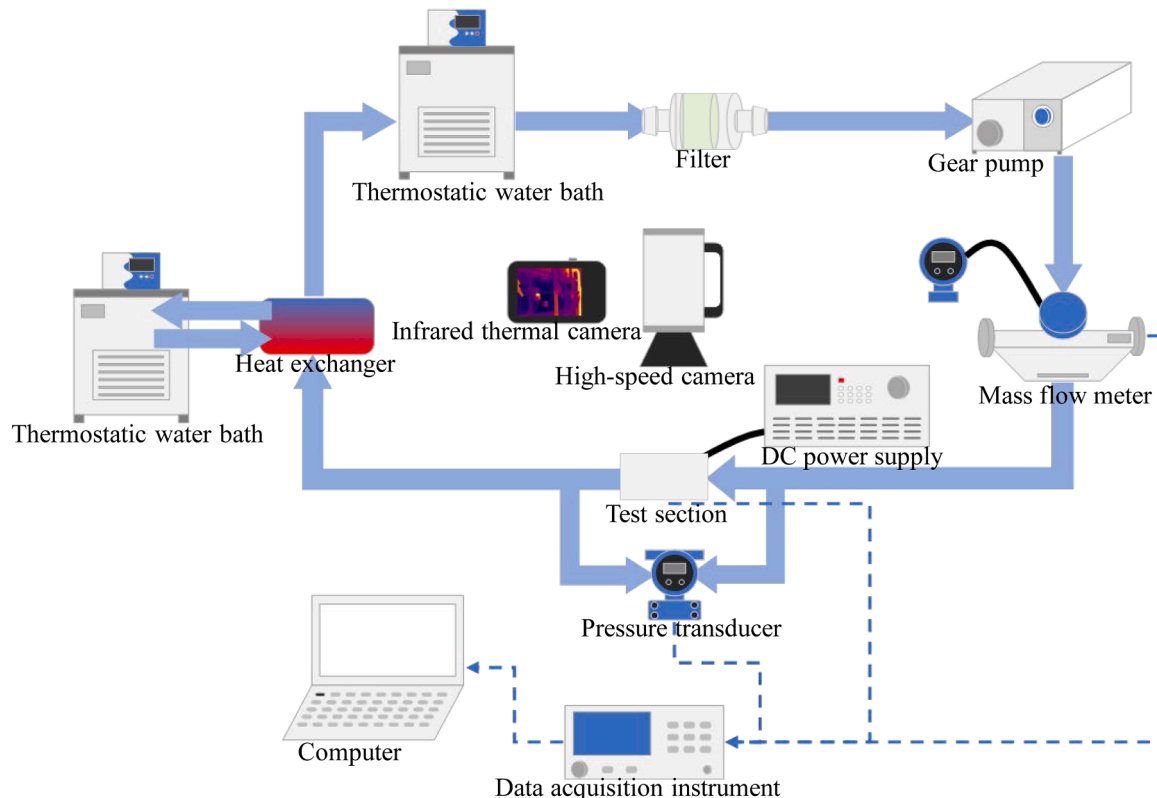


Fig. 1. Schematic of the visualized MMC heat transfer experimental setup.



Fig. 2. Experimental setup of the MMC heat transfer system.

Table 1
Specifications of the experimental setup equipment.

Equipment	Brand	Model
Thermostatic water bath	Yinggong	DC-2020
Gear pump	Cole-Parmer	07,555-05
Mass flow meter	Micro Motion	F025S113C2BPMZZZZ
DC power supply	Faith	FTP065-40-240
Pressure transducer	Yokogawa	EJA110E
Thermocouple	WSK	TT-T-36
High-speed camera	Photron	Nova s6
Infrared thermal camera	Hikmicro	HM-TPK20-3AQF/W
Data acquisition instrument	Keysight	DAQ970A

the same root channel is 5 mm. The working fluid is first distributed by the manifold channels. It then undergoes secondary distribution through the distribution plate before entering the microchannels. The introduction of this plate induces localized counter-current flow in adjacent channels. For either the odd-numbered or even-numbered channels, the manifold itself retains a two-inlet and two-outlet configuration. From this perspective, it is comparable to the two-inlet and two-outlet manifold arrangement in the parallel-flow manifold microchannel. To prevent flow stagnation at the channel ends and the consequent degradation of heat transfer performance, the methodology from previous studies is therefore adopted [38]. The microchannel structure employed by Li et al. [34] is adopted. The specific dimensions of the flow distribution plate and a schematic of the internal fluid flow within the CFMMC are presented in Fig. 4. Fig. 5 provides the photograph of the test section. After fabrication, the channel depth is measured using a depth gauge. The measured depth of the finished microchannel is approximately 1.025 mm. The fabrication error is 2.5 %. It indicating high manufacturing accuracy. The fabricated monolithic copper microchannel and distribution plate are measured using a handheld microscope. The measured channel width is approximately 0.51 mm. The fabrication error is 2 %. The measured length of the distribution plate holes is approximately 2.05 mm. The fabrication error is 2.5 %.

The results indicate high fabrication accuracy. After the test section is assembled, the heating block is wrapped with insulation wool. Any gaps between the insulation wool and the heating block are filled with a high-temperature alumina silicate ceramic fiber blanket. This minimizes heat loss to the greatest extent possible. It ensures the accuracy of the experimental data.

2.3. Data reduction and uncertainty analysis

The total heat power is the heat input by the heating rods. It is calculated as follows:

$$Q_{\text{tot}} = UI \quad (1)$$

Where U represents the input voltage of the heating rods, and I represents the input current of the heating rods.

The effective heat power under single-phase conditions is calculated as follows:

$$Q_{\text{eff}} = \dot{q}_m c_p (T_{f,\text{out}} - T_{f,\text{in}}) \quad (2)$$

Where $\dot{q}_m$ is the working fluid mass flow rate, c_p is the specific heat capacity, $T_{f,\text{out}}$ and $T_{f,\text{in}}$ represent the outlet and inlet temperatures of the working fluid, respectively.

Although insulation materials were applied to the test section, heat loss remained unavoidable and can be calculated as:

$$Q_{\text{loss}} = Q_{\text{tot}} - Q_{\text{eff}} \quad (3)$$

The effective heat flux q_{eff} can be calculated by:

$$q_{\text{eff}} = \frac{Q_{\text{eff}}}{A_b} \quad (4)$$

Where A_b is the base area of the heat sink.

Assuming that the heat transfer between measuring points and microchannel base is a one-dimensional heat conduction, the average temperature of heat sink base $T_{b,\text{ave}}$ is calculated according to Fourier's

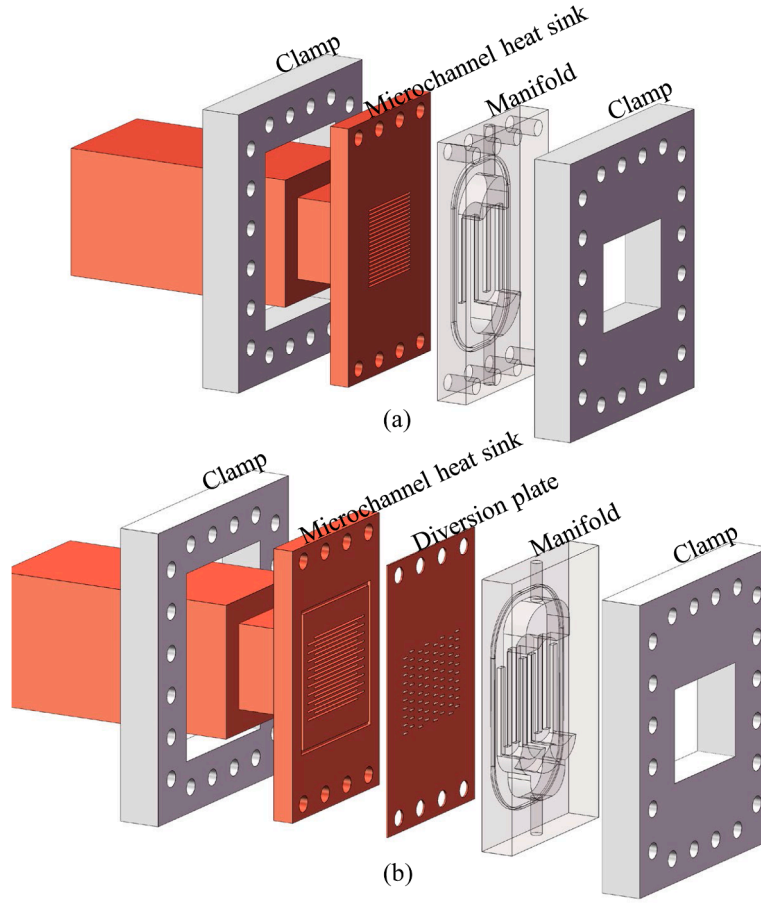


Fig. 3. Schematic diagram of the experimental test section: (a) PFMMC; (b) CFMMC.

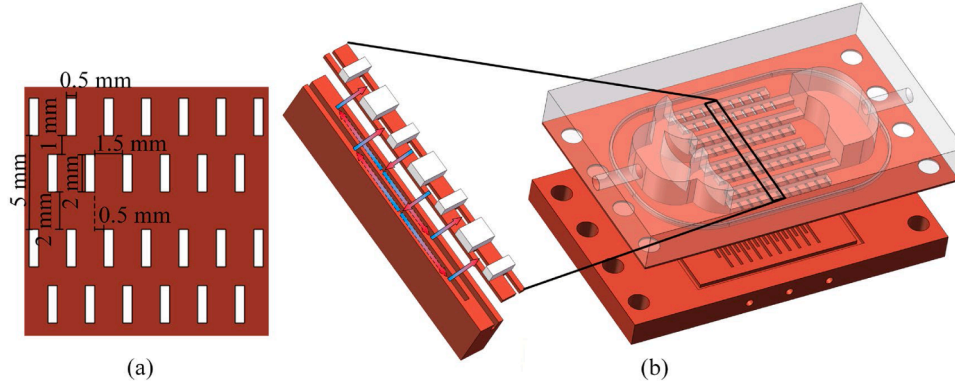


Fig. 4. (a) Dimensions of the flow distribution plate; (b) Schematic of the internal fluid flow in the CFMMC.

law as:

$$T_{b,ave} = T_{p,ave} - q_{eff} \times \frac{\Delta x}{k} \quad (5)$$

$$T_{p,ave} = \frac{T_1 + T_2 + T_3}{3} \quad (6)$$

Where $T_{p,ave}$ is the average temperature of three measuring points at the bottom of the test section, Δx is the distance between the thermocouples and the microchannel base, k is the thermal conductivity of copper.

The thermal resistance is calculated as follows:

$$R_t = \frac{T_{b,ave} - T_{f,in}}{q_{eff}} \quad (7)$$

The average convective heat transfer coefficient is calculated as follows:

$$h_{ave} = \frac{q_{eff} A_b}{A_t (T_{b,ave} - T_f)} \quad (8)$$

A_t is the total heat transfer area of the microchannel, which can be calculated by:

$$A_t = A_b - A_{b,fin} + \eta n_{fin} A_{fin} \quad (9)$$

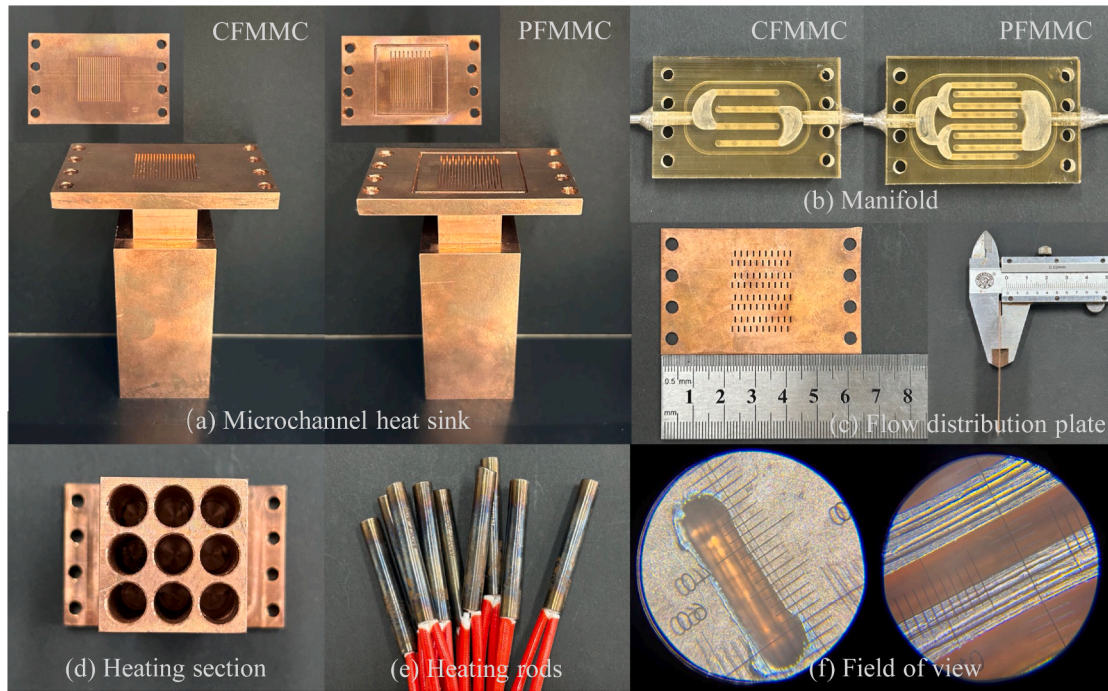


Fig. 5. Photograph of the test section: (a) Microchannel heat sink; (b) Manifold; (c) Flow distribution plate; (d) Heating section; (e) Heating rods; (f) Field of view.

Where $A_{b,fin}$ is the base area of fins, η is the fin efficiency, n_{fin} is the number of fins, and A_{fin} is the wetted area of a fin.

T_f is the reference temperature of working fluid, which can be calculated by:

$$T_f = \frac{T_{f,in} + T_{f,out}}{2} \quad (10)$$

Fin efficiency can be calculated by:

$$\eta = \frac{\tanh(mH_{fin})}{mH_{fin}} \quad (11)$$

$$m = \sqrt{\frac{P_{fin}h_{ave}}{kA_{c,fin}}} \quad (12)$$

Where m is the fin parameter, H_{fin} is the height of the fin, P_{fin} is the perimeter of the fin, and $A_{c,fin}$ is the cross-sectional areas of the fin.

Performance evaluation criterion (PEC) can be calculated by:

$$Nu = \frac{h_{ave}D_h}{k} \quad (13)$$

$$f = \frac{2\Delta PD_h}{\rho_f \nu_f^2 l_c} \quad (14)$$

$$PEC = \frac{Nu/Nu_0}{(f/f_0)^{1/3}} \quad (15)$$

Where Nu is the Nusselt number, D_h is the hydraulic diameter of the channel, ν_f is the working fluid velocity, l_c is the channel length, f is the friction factor.

The coefficient of performance (COP) is defined as the ratio of effective heat power to pumping power. It is calculated as follows

$$COP = \frac{q_{eff}A_b}{\Delta PG_v} \quad (16)$$

Where G_v is the volumetric flow rate of the working fluid.

An uncertainty analysis is performed to ensure data reliability.

Table 2

Uncertainties of the experimental parameters.

Parameters	Values
Temperature	$\pm 0.2\%$
Pressure drop	$\pm 0.3\%$
Flow rate	$\pm 0.2\%$
Heat flux	$\pm 2.8\%$
Heat transfer coefficient	$\pm 3.9\%$
COP	$\pm 4.9\%$
PEC	$\pm 5.6\%$

Table 2 presents the uncertainties of the experimental parameters. The uncertainties of directly measured parameters are determined based on the errors of their respective sensors. The uncertainties of indirect parameters are calculated using the method proposed by Moffat [40], as follows:

$$\delta R = \left(\sum_{i=1}^n \left(\frac{\partial R}{\partial V_i} \delta V_i \right)^2 \right)^{\frac{1}{2}} \quad (17)$$

3. Numerical simulation

3.1. Physical model

Fig. 6 shows the models of PFMMC and CFMMC used in the numerical simulation. The bolt holes on the monolithic copper microchannels and the manifold are located outside the main fluid flow path. As they are not directly involved in the heat transfer process, their impact on the flow and heat transfer performance is negligible. However, they significantly increase the complexity of the mesh in the simulation model. Therefore, to simplify the simulation, structures such as bolt holes and sealing grooves are filled. The heat source in the heating block is simplified as a rectangular region measuring $20 \text{ mm} \times 20.5 \text{ mm}$. In PFMMC, fluid enters the manifold inlet reservoir from the left inlet pipe. And then distributed by the manifold into the microchannels. After flowing through the channels, the fluid returns to the manifold outlet reservoir and exits through the outlet pipe. In the

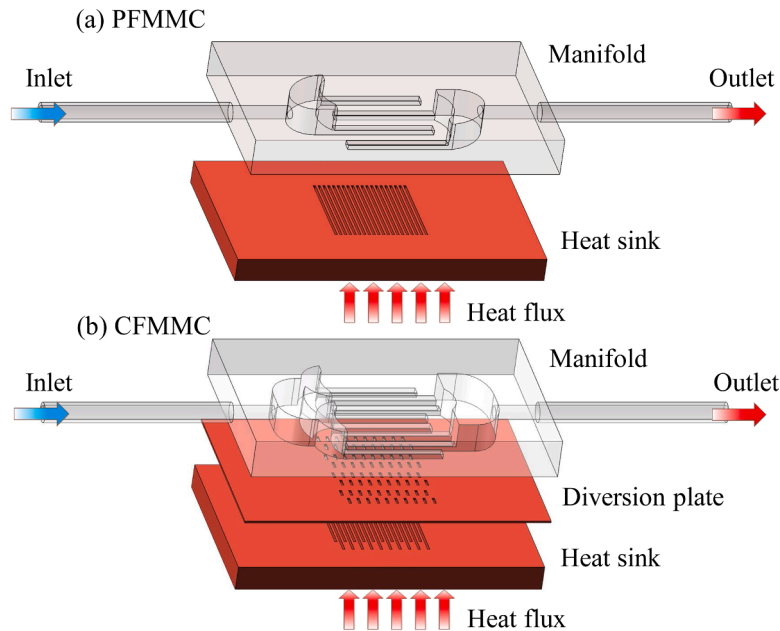


Fig. 6. Models used in the numerical simulation: (a) PFMMC; (b) CFMMC.

CFMMC, the fluid is first distributed by the manifold. It then undergoes secondary distribution through the flow distribution plate before entering the microchannels. After flowing through the channels, the fluid returns to the manifold outlet reservoir and exits through the outlet pipe.

3.2. Governing equations

To simplify the simulation, several assumptions are made for the single-phase flow. The fluid flow is assumed to be laminar. Both the heat transfer and fluid flow processes are assumed to be steady-state. Body forces and surface tension are neglected. Based on these assumptions, the fluid flow and heat transfer are governed by the following equations:

Continuity equation:

$$\nabla \cdot \vec{v} = 0 \quad (18)$$

Momentum equation:

$$(\nabla \cdot \vec{v}) \rho_f \vec{v} = -\nabla P + \mu \nabla^2 \vec{v} \quad (19)$$

Energy equation:

$$\rho_f c_{p,l} (\vec{v} \cdot \nabla T) = k_f \nabla^2 T \quad (20)$$

Where $\vec{v}$ is the velocity vector (m/s), ρ is the fluid density (kg/m³), P is the pressure

(Pa), μ is the dynamic viscosity (Pa·s), T is the fluid temperature (K), c_p is the specific heat capacity (J/(kg·K)), k is the fluid thermal conductivity (W/(m·K)).

3.3. Boundary conditions

Copper is assigned as the material for the heat sink region. Polysulfone (PSU) is specified for the manifold region. The inlet and outlet pipes are stainless steel. Water serves as the coolant. Numerical simulations are performed using ANSYS Fluent 2023 R1. The working fluid enters the manifold microchannel heat sink at a steady state with a constant mass flow rate. The inlet boundary condition is set as a mass-flow inlet. The inlet temperature is 298.15 K. The flow rate range is from 80 g/min to 170 g/min. After calculation, the Reynolds numbers within this flow range are all within the laminar flow range, and the

assumption of laminar flow is valid. The outlet boundary condition is set as a pressure outlet. A constant heat flux is applied to the heat source surface. The heat flux range is from 50 W/cm² to 70 W/cm². All other surfaces are adiabatic. The pressure-velocity coupling is solved using the Coupled scheme. The discretization of momentum and energy equations is performed with the second-order upwind scheme. The material properties used in the simulation are listed in Table 3.

3.4. Grid independence study

A grid independence study is conducted to ensure the results are independent of mesh density. To improve computational efficiency, five sets of grids with different numbers are generated. The total grids are 1.1×10^6 , 4.8×10^6 , 5.7×10^6 , 9.7×10^6 , and 1.2×10^7 . Fig. 7 shows the variations in the average temperature of the heating surface and the pressure drop across the microchannel heat sink with the number of grids. The average temperature of the heating surface decreases as the grid number increases. Conversely, the pressure drop rises. When the grid number increases from 9.7×10^6 to 1.2×10^7 , the relative error in the average heating surface temperature is 0.89 %. The relative error in pressure drop is 1.08 %. These results are presented in Table 4. To balance numerical accuracy with reasonable computational time, the mesh with 9.7×10^6 cells is selected for the computational domain.

3.5. Numerical simulation verification

To validate the reliability of the numerical methodology, the simulation results for the PFMMC are compared with the experimental data. The inlet temperature is 298.15 K. The mass flow rate ranges from 80 g/min to 170 g/min. The effective heat flux is 46.76 W/cm². As shown in Fig. 8, the trends of the simulation results agree with the experimental data. The differences between them are small. The maximum deviation

Table 3
Physical property Parameters of Materials.

	Copper	PSU	Stainless steel	Water
ρ [kg/m ³]	8978	1240	8030	998.2
μ [kg/(m·s)]	-	-	-	0.001003
C_p [J/(kg·K)]	381	1255	502.48	4182
k [W/(m·K)]	387.6	0.26	16.27	0.6

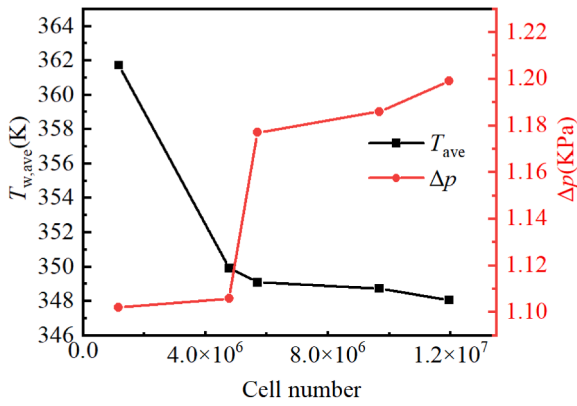


Fig. 7. Grid independence study.

Table 4

Relative errors of average heating surface temperature and pressure drop for different grid numbers.

Grid number	$T_{w,ave}$ (K)	Relative error	P (kPa)	Relative error
1.1×10^6	361.728	18.21 %	1.102	8.09 %
4.8×10^6	349.925	2.46 %	1.106	7.76 %
5.7×10^6	349.095	1.35 %	1.177	1.84 %
9.7×10^6	348.747	0.89 %	1.186	1.08 %
1.2×10^7	348.081	-	1.199	-

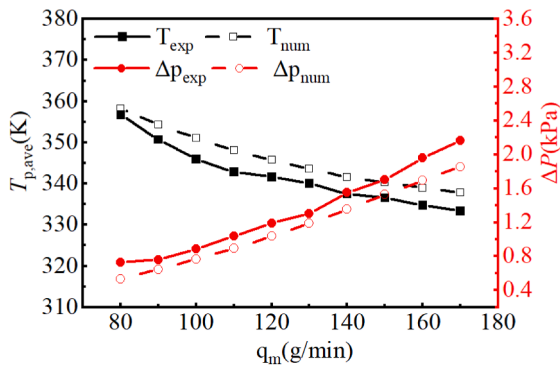


Fig. 8. Numerical simulation verification.

of the measured point temperature is 7.5 %. The mean relative error is 5.9 %. The mean relative error for the inlet-outlet pressure drop is 14.25 %. Overall, the numerical results show good agreement with the

experimental data.

4. Result analysis and discussion

4.1. Comprehensive performance analysis

Fig. 9 shows the variations of COP and PEC for the parallel-flow manifold microchannel heat sink (PFMMC) and the counter-flow manifold microchannel heat sink (CFMMC) with the mass flow rate of the working fluid under three heat fluxes during the experiment. COP quantifies the relationship between pumping power and effective heat power. A higher COP indicates better heat exchange performance per unit of pumping power. As shown in the figure, under the same heat flux, the COP for both CFMMC and PFMMC decreases significantly as the mass flow rate increases. For CFMMC, the COP decreases by approximately 91.52 % when the flow rate increases from 80 g/min to 170 g/min. This occurs because the rate of increase in pumping power exceeds the rate of improvement in heat transfer performance. Consequently, the energy utilization efficiency declines. At the same mass flow rate, a higher heat flux results in a higher COP for both CFMMC and PFMMC. This is because a greater heat flux enables the unit mass of working fluid to remove more heat. Under identical heat flux conditions, the COP of CFMMC is higher than that of PFMMC. The difference in COP between the two configurations decreases significantly as the mass flow rate increases. At a total heat flux of 50 W/cm² and a mass flow rate of 80 g/min, the COP of CFMMC is approximately 130 % higher than that of PFMMC. At a mass flow rate of 170 g/min, the COP of CFMMC is approximately 32 % higher than that of PFMMC.

PEC is a dimensionless parameter that considers both flow resistance and heat transfer performance. In this study, the PEC value of PFMMC at a total heat flux of 50 W/cm² is selected as the baseline reference. A PEC greater than 1 indicates superior comprehensive performance compared to the baseline. A higher PEC value corresponds to better comprehensive performance. At lower mass flow rates, such as 80 g/min to 100 g/min, the PEC gradually decreases as the flow rate increases. When the mass flow rate exceeds 100 g/min, the PEC stabilizes and exhibits only minor fluctuations. This occurs because, at low flow rates, the increase in flow resistance outpaces the improvement in heat transfer performance, leading to a decline in comprehensive performance. At high flow rates, a dynamic balance between heat transfer enhancement and flow resistance is achieved, resulting in a stabilized PEC with only slight variations. Under the three heat flux conditions, the PEC of PFMMC remains mostly below 1.5, approaching 1 under some operating conditions. In contrast, the PEC of CFMMC is predominantly above 1.4 and shows greater stability at high flow rates. This indicates that CFMMC exhibits a more significant advantage in comprehensive heat transfer performance. At the same mass flow rate, a higher heat flux leads to a higher

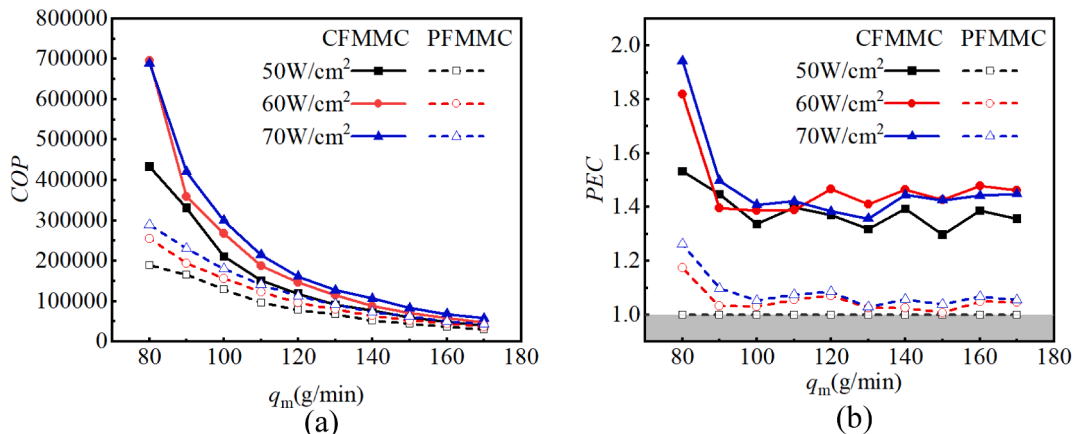


Fig. 9. Variations in COP and PEC with working fluid mass flow rate.

PEC. This is because an increased heat flux raises the channel wall temperature. Based on the physical analysis of the experimental phenomena, for the working fluid, the dynamic viscosity decreases with increasing temperature. The reduction in viscosity results in a thinner velocity boundary layer and weaker viscous forces, which reduces flow resistance. Simultaneously, the thermal boundary layer becomes thinner concurrently. The thermal boundary layer is the primary thermal resistance to heat transfer. Its thinning significantly reduces conductive thermal resistance. Consequently, the convective heat transfer coefficient is substantially enhanced, and the Nusselt number increases markedly. This leads to a corresponding increase in PEC.

4.2. Flow characteristics analysis

Fig. 10 presents the simulated velocity contours for CFMMC and PFMMC. The conditions correspond to the experiment with a total heat

flux of 50 W/cm^2 and a working fluid mass flow rate of 100 g/min . Each cross-sectional contour is taken from a slice at the manifold channel centerline. The slice extends from the top surface of the manifold channel down to the measurement plane beneath the microchannels. The simulation results indicate that CFMMC exhibits a more uniform velocity distribution. Due to the impingement of the working fluid on the manifold reservoir, both configurations show higher fluid velocities in the manifold reservoir near the inlet pipe. The cross-sectional contours reveal that the fluid velocity in the manifold inlet channels gradually decreases along the flow direction. This occurs because a gradually narrowing structure exists between the inlet channels and the reservoir. This geometry leads to higher fluid velocities at the interface between the manifold channels and the manifold reservoir. Conversely, the velocity in the manifold outlet channels increases slightly.

Fig. 11 shows the variation of mass fluxes in each channel of CFMMC and PFMMC along the direction of working fluid flow under the

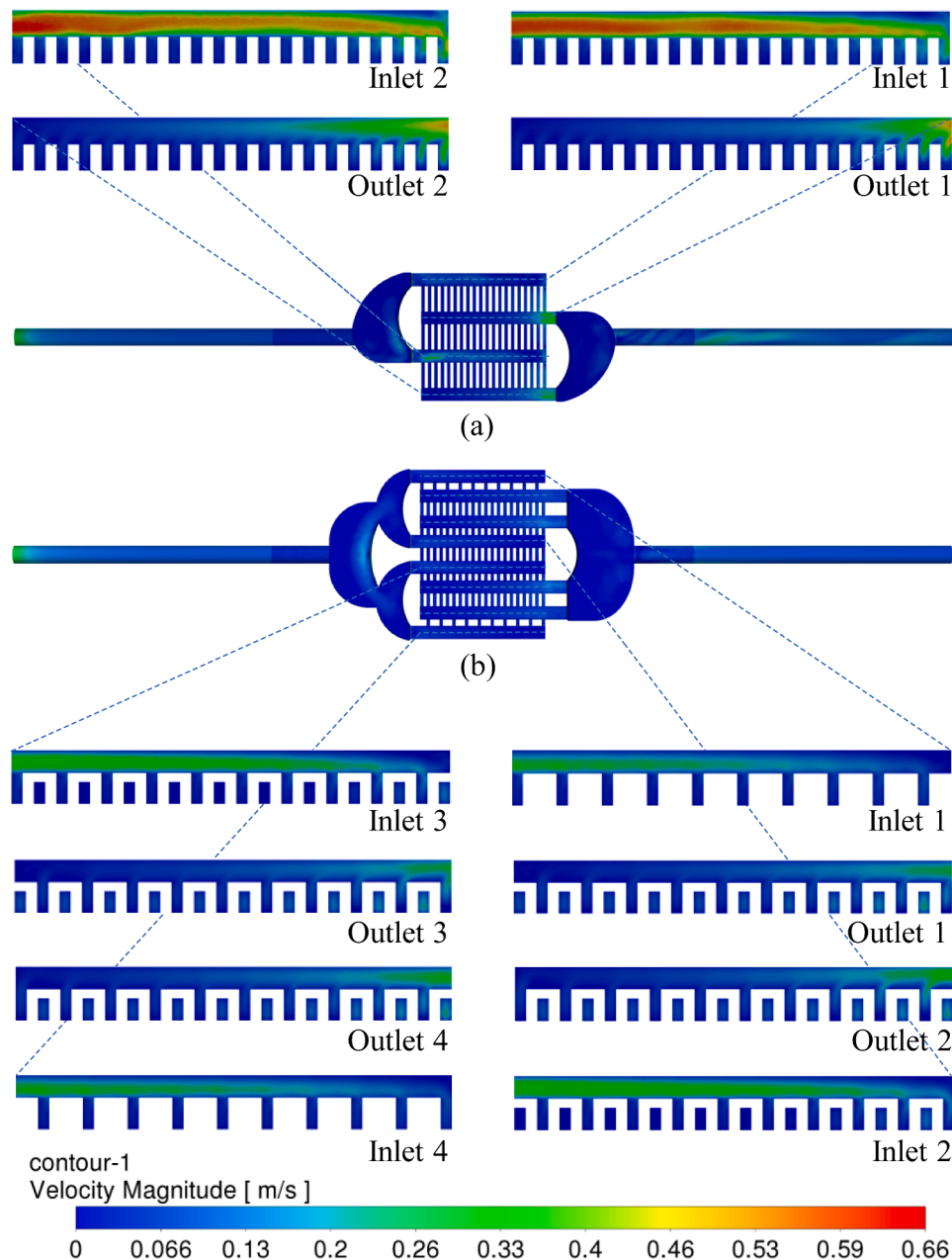


Fig. 10. Velocity contours for the overall manifold microchannel heat sinks and the longitudinal cross-section along the centerline of the manifold flow channels: (a) PFMMC, (b) CFMMC.

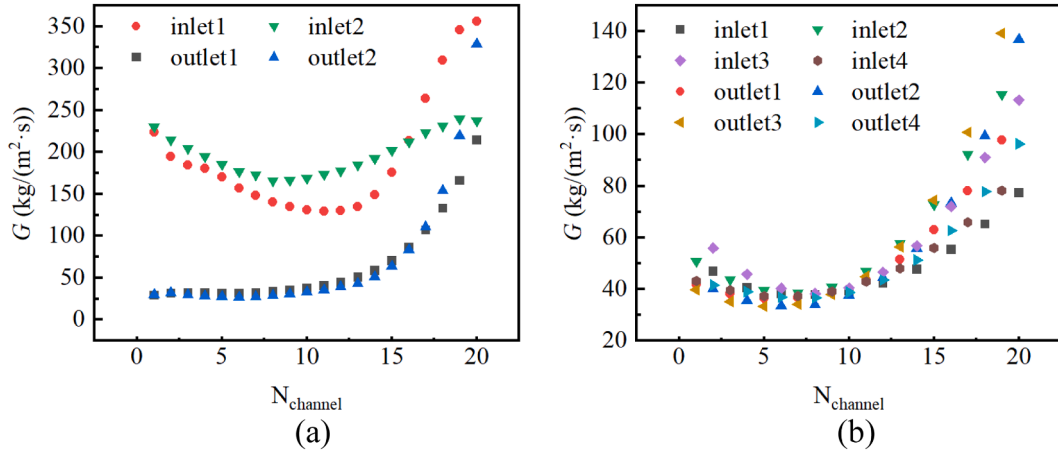


Fig. 11. The variation of mass fluxes in each channel: (a) PFMMC (b) CFMMC.

conditions of a total heat flux of 50 W/cm^2 and a working fluid mass flow rate of 100 g/min . Owing to its two-inlet and two-outlet manifold configuration, PFMMC exhibits an overall higher mass flux within the channels. The mass flux fluctuates within a range of $26 \text{ kg/m}^2/\text{s}$ to $357 \text{ kg/m}^2/\text{s}$. For PFMMC, the mass flux at both inlets initially decreases and then increases along the flow direction. The inlet mass flux values are also higher than those at the two outlets. Specifically, the inlet closer to the water inlet pipe shows a higher mass flux than the inlet farther from the inlet pipe. In contrast, CFMMC employs a four-inlet and four-outlet manifold configuration. For the same total inlet flow rate, this configuration achieves more uniform flow distribution within the channels. Consequently, the mass flux is relatively lower and fluctuates within a narrower range, from $33 \text{ kg/m}^2/\text{s}$ to $140 \text{ kg/m}^2/\text{s}$. The minimum mass flux within the channels is higher, while the maximum mass flux is lower. Compared to the PFMMC, the minimum mass flux within the channels increases by approximately 27 %, while the maximum mass flux decreases by approximately 61 %. For CFMMC, the difference between the inlet and outlet mass fluxes is small. Both also exhibit a trend of first decreasing and then increasing along the flow direction.

To better characterize the flow distribution in the two structures, two evaluation metrics are introduced: the maximum maldistribution index and the maldistribution index. The maximum maldistribution index is defined as the ratio of the difference between the maximum and minimum mass fluxes to the average mass flux across all channels. This parameter captures the most extreme flow maldistribution, and its calculation is given by Eq. (21). The maldistribution index is calculated based on the relative standard deviation of the mass flux in all channels,

reflecting the overall dispersion relative to the mean. Its formulation is presented in Eq. (22). Fig. 12 shows the variations of the maximum maldistribution index and the maldistribution index with mass flow rate for both structures under a total heat flux of 50 W/cm^2 . As shown in Fig. 12 (a), at a mass flow rate of 100 g/min , the maximum maldistribution index is 249.52 for the PFMMC and 191.83 for the CFMMC. The CFMMC exhibits a 23.11 % reduction in the maximum maldistribution index. This demonstrates that the introduction of the flow distribution plate and the four-inlet and four-outlet manifold configuration make the working fluid flow distribution in each channel more uniform, effectively mitigating the flow differences between channels. Similarly, as shown in Fig. 12 (b), at a mass flow rate of 100 g/min , the maldistribution index is 65.88 for the PFMMC and 43.58 for the CFMMC, corresponding to a reduction of 33.85 % for the CFMMC. Compared with the maximum maldistribution index, the maldistribution index yields relatively lower values, all remaining below 100 %. This occurs because the maldistribution index incorporates the mass flux from all channels. Its value is less susceptible to domination by a single extreme value, thereby avoiding anomalously high results. In contrast, the maximum maldistribution index directly captures the most extreme maldistribution within the system, namely the difference between the channel with the highest flow rate and that with the lowest, which can lead to relatively large values. Although the numerical ranges obtained from the two calculation methods differ, both metrics consistently indicate that the CFMMC exhibits significantly better flow uniformity than the PFMMC.

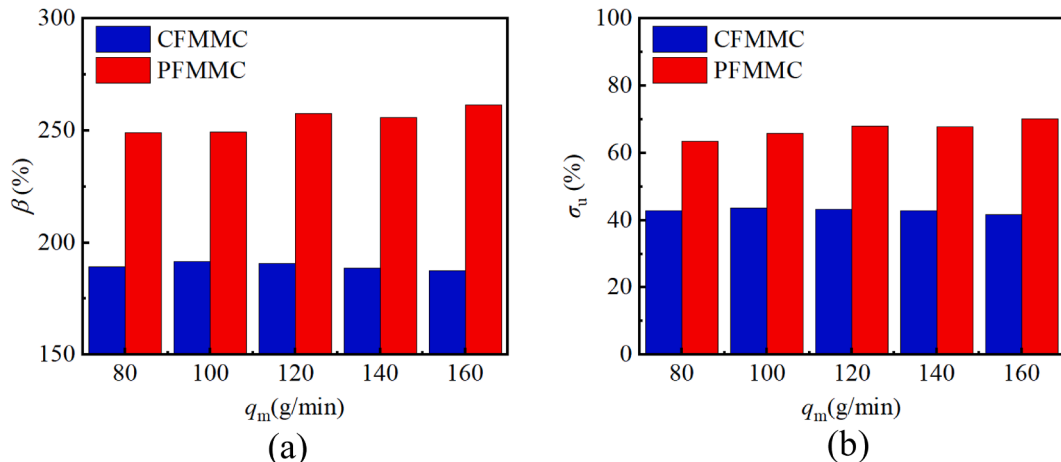


Fig. 12. (a) Maximum maldistribution index; (b) Maldistribution index.

$$\beta = \frac{G_{\max} - G_{\min}}{G_{\text{ave}}} \times 100\% \quad (21)$$

Here, β is the maximum maldistribution index, $G_{\max}$ is the maximum mass flux among all channels, $G_{\min}$ is the minimum mass flux among all channels, and G_{ave} is the average mass flux across all channels.

$$\sigma_u = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{G_c(i)}{G_{\text{ave}}} - 1 \right)^2} \times 100\% \quad (22)$$

$$G_{\text{ave}} = \frac{1}{n} \sum_{i=1}^n G_c(i), (i = 1, 2, \dots, n)$$

Here, $n = 20$, σ_u is the maldistribution index, and G_c is the mass flux of each microchannel.

Fig. 13 presents the fluid streamlines on the channel bottom surface. In PFMMC, the fluid flows in the same direction across all channels. In CFMMC, localized counter-current flow emerges in adjacent channels. This is because the distribution holes on the plate are staggered. With this plate in place, the fluid within the odd-numbered and even-numbered channels still effectively follows a two-inlet and two-outlet manifold configuration. However, due to the positional offset between the manifold inlets feeding the odd-numbered and even-numbered channels, the flow direction in adjacent channels is opposite, resulting in the counter-current flow. Since the channels closer to the outlet pipe exhibit higher mass flux, the fluid velocity in the right-side channels is relatively higher, as observable in Fig. 13.

4.3. Pressure characteristics analysis

Fig. 14 shows the variations of pressure drop and friction factor of two types of manifold microchannel heat sinks with the mass flow rate of the working fluid under three heat flux levels during the experiment. As shown in the figure, under the same heat flux, the pressure drop for both CFMMC and PFMMC increases gradually with increasing mass flow rate. At the same mass flow rate, variations in heat flux have little effect on the overall pressure drop of the manifold microchannel heat sink. At the same heat flux condition, the friction factor of PFMMC decreases gradually as the mass flow rate increases. In contrast, the friction factor of CFMMC increases gradually with increasing mass flow rate. This occurs because the flow resistance in PFMMC is primarily attributed to frictional losses along the channels. A higher mass flow rate leads to a greater flow velocity. The enhanced fluid inertia weakens viscous effects

and thins the flow boundary layer. Consequently, the friction between the fluid and the channel walls is reduced, resulting in a gradual decrease in the friction factor for PFMMC as the mass flow rate increases. In contrast, the pressure loss in CFMMC is dominated by local losses associated with the flow distribution plate. In CFMMC, the fluid exhibits a high lateral velocity at the holes of the distribution plate. To enter the channels, the fluid must decelerate sharply, converting its lateral kinetic energy into pressure energy to overcome the flow resistance within the channels. Similarly, as the fluid exits the channels, it must accelerate rapidly, converting pressure energy back into kinetic energy. This process involves significant energy dissipation. The magnitude of this local dissipation far exceeds the frictional losses along the flow path. Consequently, the friction factor for CFMMC increases gradually with increasing mass flow rate. Under all investigated conditions, the overall pressure drop and friction factor of the CFMMC are lower than those of the PFMMC. This is attributed to the flow distribution process within the CFMMC. After the initial distribution by the manifold, the fluid undergoes a secondary distribution through the staggered holes on the distribution plate before entering the microchannels. This two-stage distribution ensures more uniform flow entry into each channel, fundamentally improving flow distribution uniformity. Because CFMMC eliminates the few high-velocity channels caused by uneven flow distribution in traditional Z-type manifold, the total pressure drop at the inlet and outlet of the manifold microchannel is changed from being dominated by these extremely high local resistances to being jointly determined by the relatively uniform medium resistance of all channels. Collectively, these factors result in the lower overall pressure drop observed in the CFMMC. When the mass flow rate is 170 g/min, the pressure drop of CFMMC is approximately 23 % lower than that of PFMMC.

Fig. 15 presents the simulated pressure contours for CFMMC and PFMMC under experimental conditions: a total heat flux of 50 W/cm² and a working fluid mass flow rate of 100 g/min. The pressure contours show that for both configurations, the highest pressure occurs at the inlet pipe. The pressure then gradually decreases along the flow path. CFMMC demonstrates a lower overall pressure drop. Fig. 16 shows the variation of pressure in each channel of CFMMC and PFMMC along the direction of working fluid flow under the conditions of a total heat flux of 50 W/cm² and a working fluid mass flow rate of 100 g/min. For PFMMC, the pressure at both inlets initially decreases and then increases along the flow path. The pressure at both outlets gradually decreases. The inlet pressures for each channel are relatively uniform. For Inlet 1, the pressure difference between its first and last channels is 0.04 kPa.

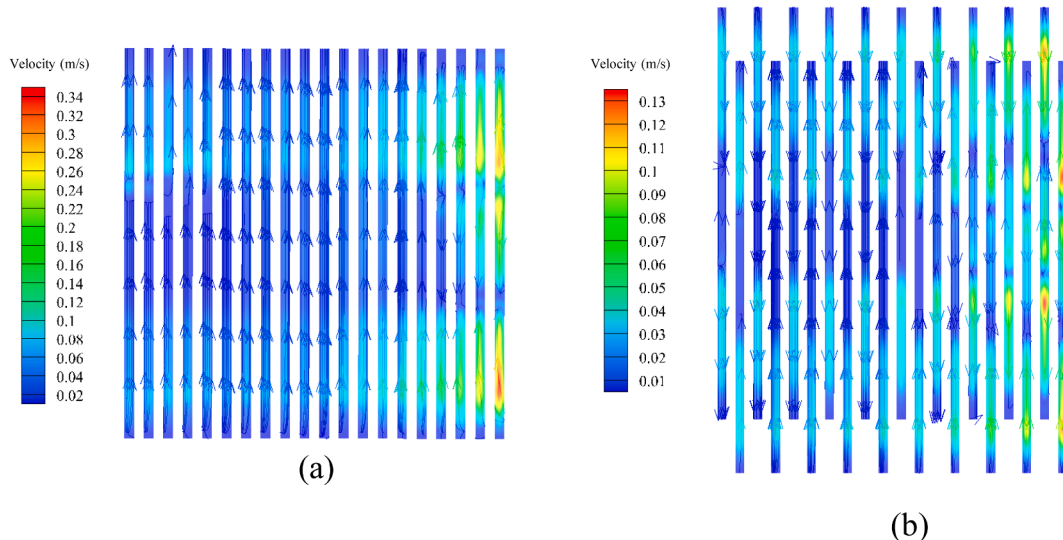


Fig. 13. Fluid streamlines on the channel bottom surface: (a) PFMMC, (b) CFMMC.

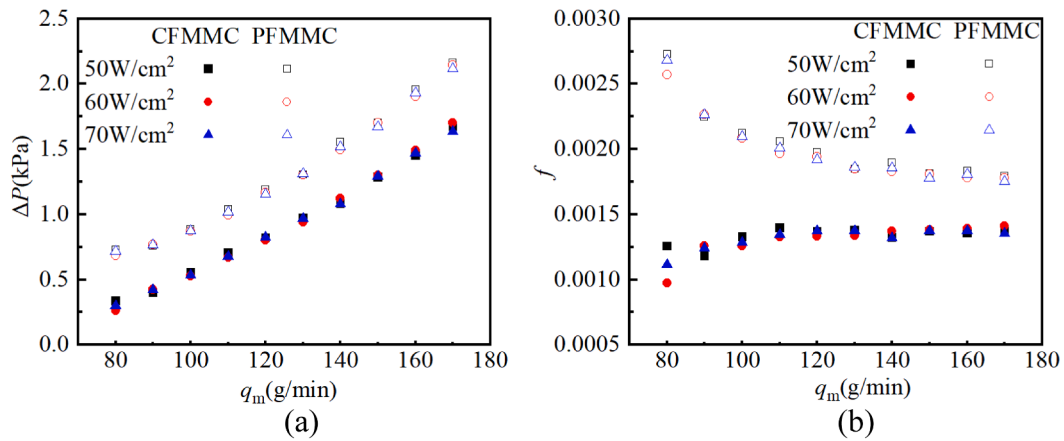


Fig. 14. Variations in pressure drop and friction factor with working fluid mass flow rate.

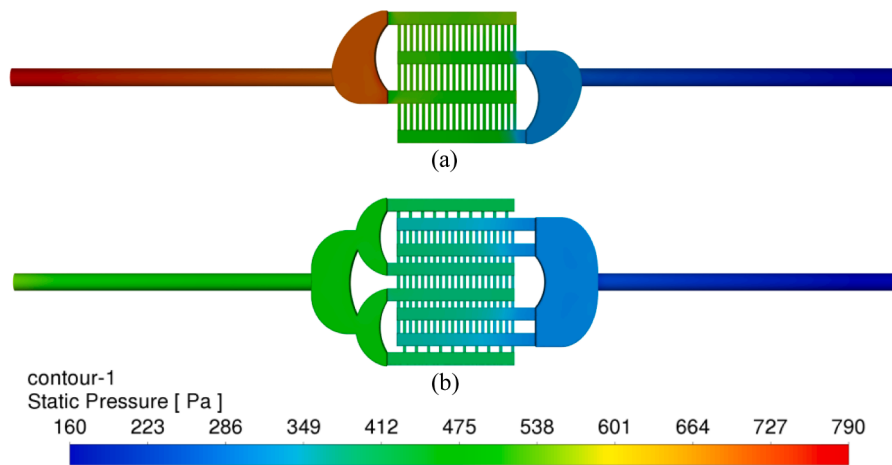


Fig. 15. Overall pressure contours of both manifold microchannel heat sinks: (a) PFMMC, (b) CFMMC.

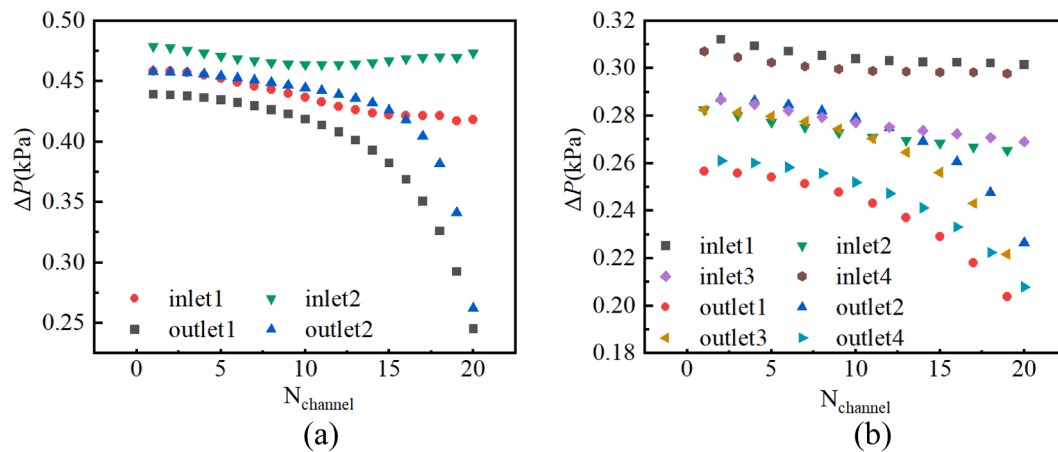


Fig. 16. The variation of pressure in each channel: (a) PFMMC (b) CFMMC.

However, the outlet pressure distribution is highly non-uniform. For Outlet 1, the pressure difference between its first and last channels is 0.194 kPa. For CFMMC, the pressures in all four inlet and four outlet channels decrease monotonically along the flow direction. For Inlet 2, the pressure difference between its first and last channels is 0.017 kPa. For Outlet 2, the corresponding pressure difference is 0.061 kPa.

4.4. Heat transfer characteristics analysis

Fig. 17 shows the variations of the channel bottom average temperature, thermal resistance, heat transfer coefficient and Nusselt number of two types of manifold microchannel heat sinks with the mass flow rate of the working fluid under three heat flux levels during the experiment. Overall, under a constant heat flux, the average channel bottom temperature and thermal resistance for both configurations

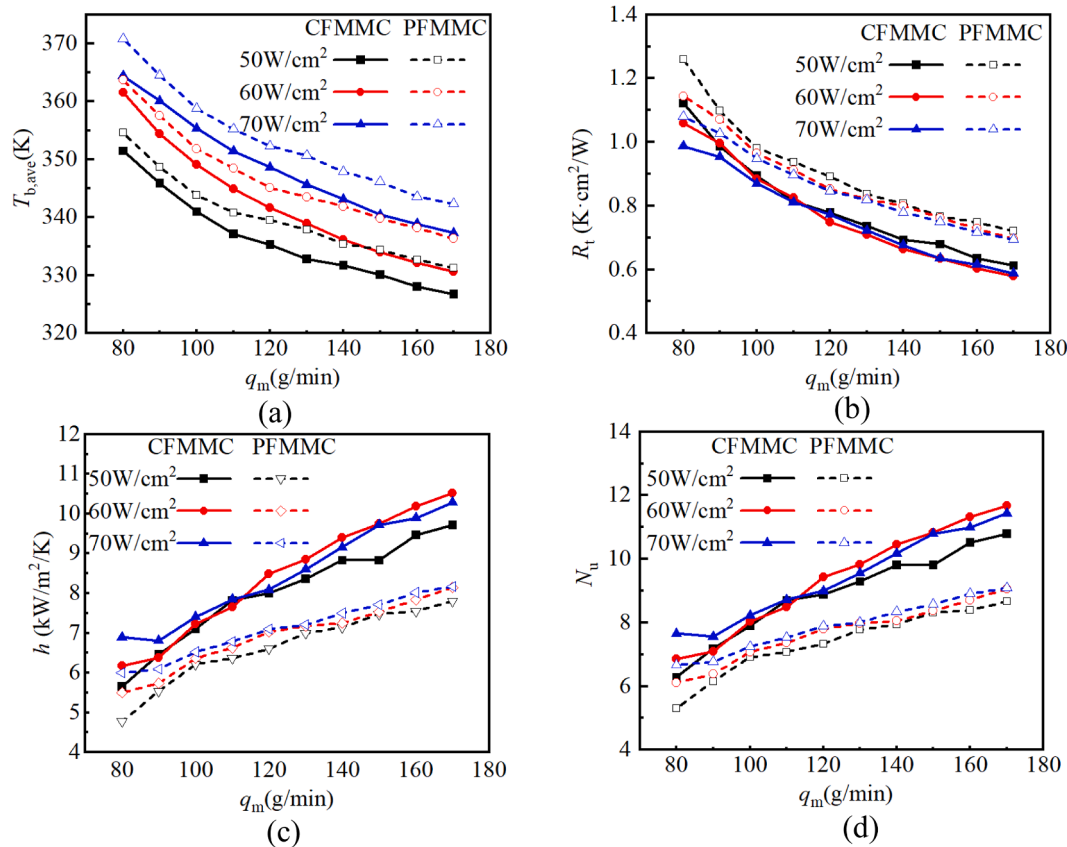


Fig. 17. The variations of the channel bottom average temperature, thermal resistance, heat transfer coefficient and Nusselt number with working fluid mass flow rate.

gradually decrease as the working fluid mass flow rate increases. Conversely, the heat transfer coefficient and Nusselt number gradually increase. When the mass flow rate is low, its variation has a significant impact on the average temperature. This occurs because, at low mass flow rates, the fluid velocity is slow. The resulting longer residence time within the channels leads to a more pronounced temperature rise of the fluid. Compared to PFMMC, under the same heat flux, CFMMC exhibits a lower average channel bottom temperature and a lower thermal resistance. At a total heat flux of 50 W/cm² and a mass flow rate of 170 g/min, compared to PFMMC, CFMMC reduces the average channel bottom temperature by approximately 4 K. The thermal resistance is reduced by about 15 %. This indicates that the CFMMC structure offers less resistance to heat transfer, facilitating faster heat dissipation. Furthermore, due to its superior heat dissipation capability and smaller temperature difference, CFMMC achieves a higher heat transfer coefficient and Nusselt number. At a total heat flux of 50 W/cm² and a mass flow rate of 170 g/min, the heat transfer coefficient for CFMMC is 9.7 kW/m²/K. In contrast, the value for PFMMC is only 7.8 kW/m²/K. This represents an enhancement of 24.4 % for CFMMC compared to PFMMC.

Fig. 18 shows the thermal images of two structures taken by an infrared camera at different flow rates when the total heat flux is 50 W/cm² during the experiment. Due to obstruction by the manifold, the infrared camera cannot precisely capture the surface temperature directly over the channels. However, the thermal images can reflect the internal flow conditions within the manifold channels through the temperature on the manifold's top surface. The thermal images show that under the same heat flux, a lower mass flow rate corresponds to a higher surface temperature on the manifold. Under identical heat flux and mass flow rate conditions, PFMMC exhibits an overall higher manifold surface temperature compared to CFMMC. Both configurations show a trend of lower temperature near the inlet side and higher

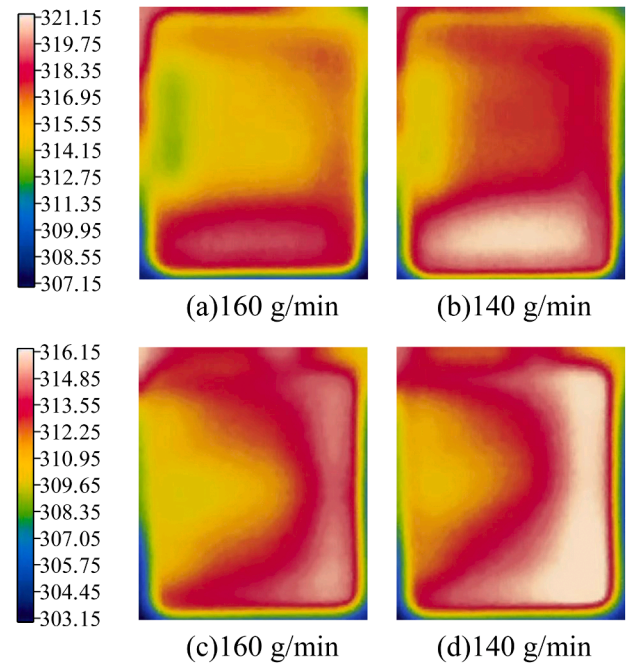


Fig. 18. Thermal images of the top surface for PFMMC and CFMMC under different flow rates: (a) PFMMC 160 g/min (b) PFMMC 140 g/min (c) CFMMC 160 g/min (d) CFMMC 140 g/min.

temperature near the outlet side. As can be seen from the previous analysis of Section 4.2, the mass flux within channels closer to the inlet

pipe is lower, while it is higher in channels farther from the inlet pipe. Consequently, less working fluid flows into the inlet-side channels, with a greater portion converging in the outlet-side channels. Meanwhile, the manifold channels on the inlet side contain a larger volume of working fluid, which is also at a relatively lower temperature. These combined effects result in the lower temperature observed on the inlet side and the higher temperature on the outlet side in the thermal images.

Fig. 19 presents the simulated temperature contours for CFMMC and PFMMC. The conditions correspond to the experiment with a total heat flux of 50 W/cm^2 and a working fluid mass flow rate of 100 g/min . Each cross-sectional contour is taken from a slice at the manifold channel centerline. The slice extends from the top surface of the manifold channel down to the measurement plane beneath the microchannels. The numerical results show that the working fluid in CFMMC has an overall lower temperature and a more uniform temperature distribution.

In contrast, PFMMC exhibits localized high-temperature zones in the manifold channels and in microchannels near the inlet. Comparing the cross-sections, the temperature of the working fluid within individual channels in CFMMC shows little variation, approximately 6 K . Furthermore, the solid domain temperature in CFMMC is lower.

Due to the limited number of temperature measurement points in the experiment, the temperature distribution on the measurement plane cannot be obtained from the experimental data alone. Numerical simulation provides detailed temperature field information, complementing the experimental approach and enabling an evaluation of the temperature uniformity of the two manifold microchannel heat sinks. Fig. 20 presents the contour map on the measurement plane where the thermocouple holes are located. The results indicate that CFMMC exhibits a lower temperature and a smaller temperature gradient on this plane. The maximum temperature on the CFMMC measurement plane is

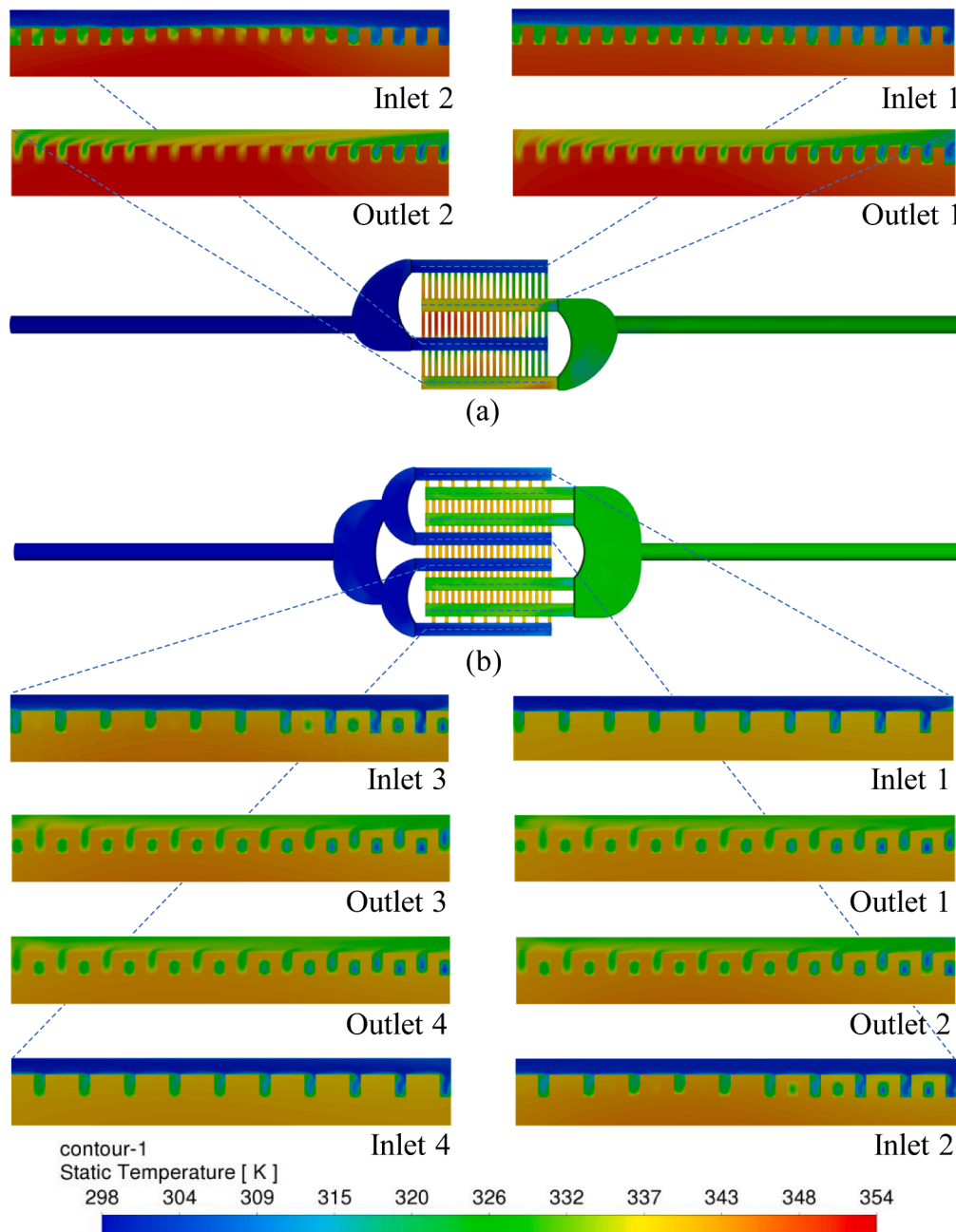


Fig. 19. Temperature contours for the overall manifold microchannel heat sinks and the longitudinal cross-section along the centerline of the manifold flow channels: (a) PFMMC, (b) CFMMC.

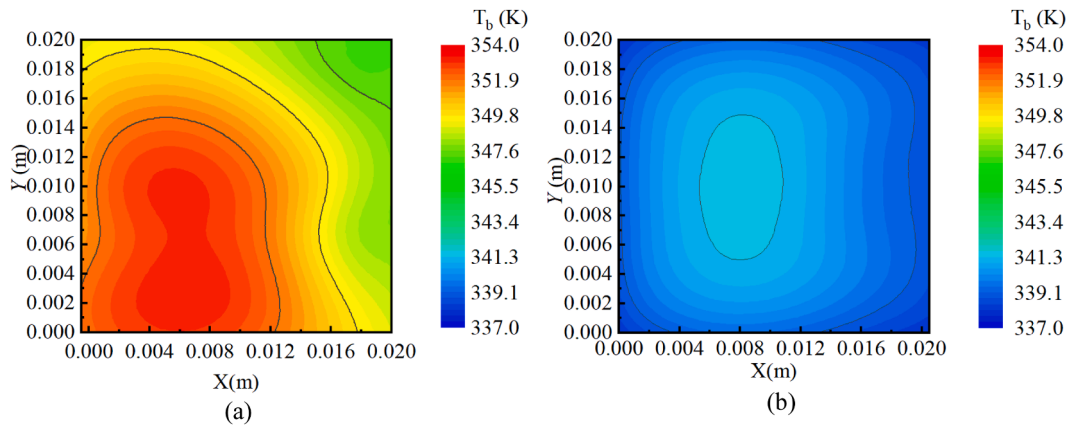


Fig. 20. Temperature contour maps: (a) PFMMC (b) CFMMC.

approximately 13 K lower than that of PFMMC, with a corresponding temperature gradient of about 4 K. For both configurations, localized high temperatures on the measurement plane appear near the inlet region, which is related to the specific structural characteristics of the models in this study. As can be seen from the previous analysis of Section 4.2, the working fluid mass flux gradually increases along the flow direction. The resulting increase in flow velocity directly enhances the heat transfer coefficient. The incoming cool working fluid from the inlet provides significant cooling to the heat sink, leading to the location of the highest temperature on the measurement plane near the inlet region.

Fig. 21 shows the variations of the maximum temperature, minimum temperature and temperature difference of the measurement plane where the thermocouple holes are located with the mass flow rate of the working fluid in numerical simulations under three heat flux levels. For both configurations, the maximum and minimum temperatures on the measurement plane decrease as the mass flow rate increases. The temperature difference for CFMMC shows an upward trend with increasing mass flow rate, while for PFMMC it exhibits a downward trend. Overall,

compared to PFMMC, CFMMC demonstrates lower maximum and minimum temperatures. At a total heat flux of 70 W/cm^2 and a mass flow rate of 170 g/min , the maximum temperature for CFMMC is approximately 15 K lower than that for PFMMC. The minimum temperature is reduced by about 12 K. Furthermore, CFMMC exhibits a smaller temperature difference. This is consistent with the previous analysis, indicating better temperature uniformity for CFMMC.

4.5. Extreme performance test

To evaluate the extreme flow and heat transfer performance of the two manifold microchannels under higher loads while ensuring laminar flow conditions, the validated numerical models are used to extend the mass flow rate up to 2000 g/min for testing. Four mass flow rates are selected: 2000 g/min , 1000 g/min , 500 g/min and 250 g/min . Fig. 22 shows the variations of the average temperature of the heating surface, the average temperature of the fluid-solid couple surface and the temperature difference of the heating surface with the effective heat flux

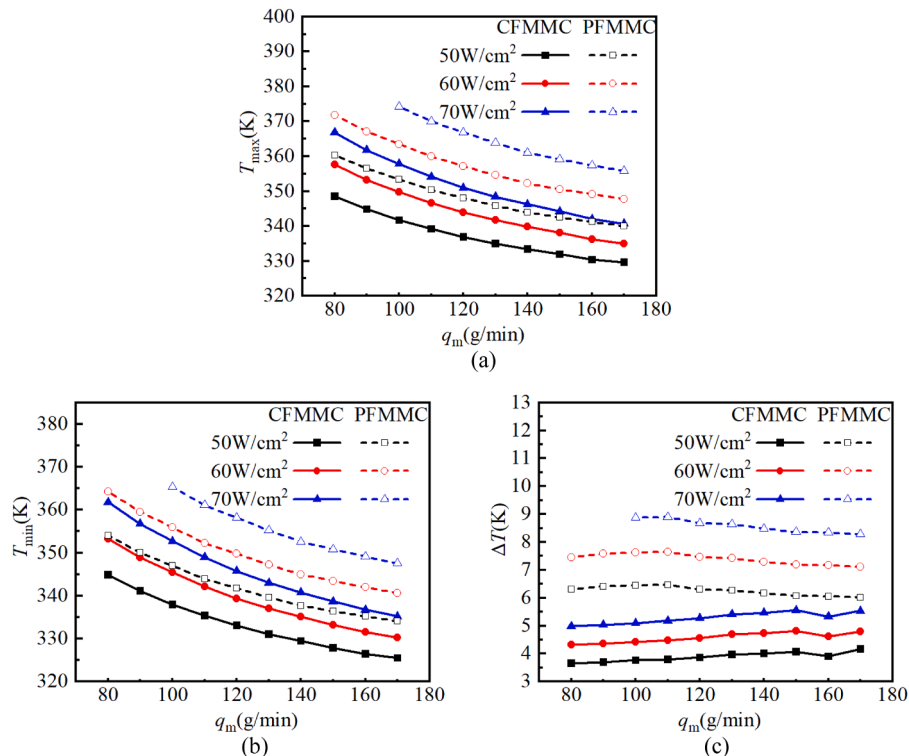


Fig. 21. The variations of the maximum temperature, minimum temperature and temperature difference with working fluid mass flow rate.

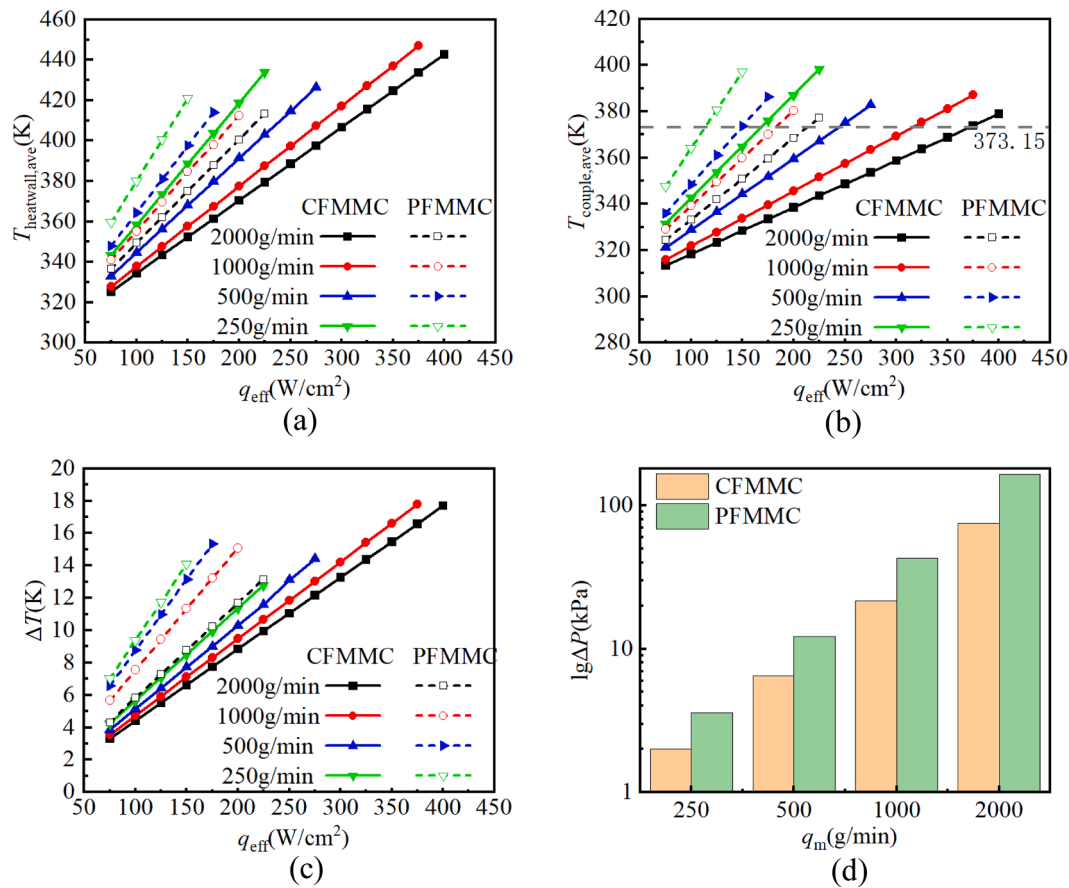


Fig. 22. (a) – (c): Variations of average heating surface temperature, average fluid-solid couple surface temperature and heating surface temperature difference with effective heat flux; (d) Variation of pressure drop with working fluid mass flow rate.

density and the pressure drop with the mass flow rate of CFMMC and PFMMC at 2000 g/min, 1000 g/min, 500 g/min and 250 g/min respectively. As shown in the figure, for both configurations, the average heating surface temperature, the average fluid-solid couple surface temperature, and the heating surface temperature difference increase with rising effective heat flux. Under the same mass flow rate and effective heat flux, CFMMC demonstrates lower values for the average heating surface temperature, the average fluid-solid couple surface temperature, and the heating surface temperature difference compared to PFMMC. At a mass flow rate of 2000 g/min under single-phase conditions, CFMMC can withstand an effective heat flux of 300 W/cm^2 , with a heating surface temperature difference of approximately 13 K. In contrast, PFMMC can withstand a maximum effective heat flux of only 175 W/cm^2 . As the flow rate increases, the difference in pressure drop between the two manifold microchannel configurations becomes more pronounced. The unique counter-flow configuration of CFMMC contributes to mitigating the flow maldistribution issue typical of Z-type manifold. At a mass flow rate of 2000 g/min, the pressure drop for CFMMC is approximately 75 kPa. This represents a reduction of about 54 % compared to PFMMC. It should be noted that the operating conditions involved in the extreme performance tests were conducted using numerical simulations and extend beyond the range of the current experimental setup (80 - 170 g/min, 50 - 70 W/cm^2). These results primarily serve to reveal the potential performance trends and theoretical limits of the CFMMC under high thermal loads. Practical application requires further validation based on specific system conditions.

5. Conclusion

To evaluate the flow and thermal performance of the CFMMC, this

study employs a manifold microchannel flow visualization experimental platform. Deionized water is used as the working fluid. Tests are conducted at total heat fluxes of 50, 60, and 70 W/cm^2 with a constant inlet temperature of 298.15 K. The flow and heat transfer characteristics of the CFMMC are compared with the PFMMC over a mass flow rate range of 80 g/min to 170 g/min. The main conclusions are as follows:

- (1) Under the same heat flux, the COP of CFMMC is higher than that of PFMMC. The difference in COP between the two configurations decreases significantly as the mass flow rate increases. At a total heat flux of 50 W/cm^2 and a mass flow rate of 80 g/min, the COP of CFMMC is approximately 130 % higher than that of PFMMC. Under the three heat flux conditions, the PEC of PFMMC remains mostly below 1.5, approaching 1 under some operating conditions. In contrast, the PEC of CFMMC is predominantly above 1.4 and shows greater stability at high flow rates. This indicates that CFMMC exhibits a more significant advantage in comprehensive heat transfer performance.
- (2) At a total heat flux of 50 W/cm^2 and a mass flow rate of 100 g/min, the mass flux fluctuates within a range of 26 $\text{kg}/\text{m}^2/\text{s}$ to 357 $\text{kg}/\text{m}^2/\text{s}$. For the same total inlet flow rate, CFMMC achieves more uniform flow distribution within the channels. Consequently, the mass flux is relatively lower and fluctuates within a narrower range, from 33 $\text{kg}/\text{m}^2/\text{s}$ to 140 $\text{kg}/\text{m}^2/\text{s}$. The minimum mass flux within the channels is higher, while the maximum mass flux is lower. The maximum maldistribution index is 249.52 for the PFMMC and 191.83 for the CFMMC. The CFMMC exhibits a 23.11 % reduction in the maximum maldistribution index.
- (3) The introduction of the flow distribution plate and the four-inlet and four-outlet manifold configuration leads to more uniform

working fluid distribution among the individual channels. Under all tested operating conditions, CFMMC exhibits lower overall pressure drop and friction factor than PFMMC. When the mass flow rate is 170 g/min, the pressure drop of CFMMC is approximately 23 % lower than that of PFMMC.

- (4) Compared to PFMMC, under the same heat flux, CFMMC exhibits a lower average channel bottom temperature and a lower thermal resistance. Furthermore, due to its superior heat dissipation capability and smaller temperature difference, CFMMC achieves a higher heat transfer coefficient and Nusselt number. At a total heat flux of 50 W/cm² and a mass flow rate of 170 g/min, compared to PFMMC, CFMMC reduces the average channel bottom temperature by approximately 4 K. The thermal resistance is reduced by about 15 % and the heat transfer coefficient is improved by about 24.4 %.
- (5) Numerical simulations are used to test the flow and heat transfer performance of both manifold microchannel heat sinks. At a mass flow rate of 2000 g/min under single-phase conditions, PFMMC can withstand a maximum effective heat flux of 175 W/cm². In contrast, CFMMC can withstand an effective heat flux of 300 W/cm² while simultaneously reducing the pressure drop by approximately 54 % compared to PFMMC.
- (6) Within the experimental ranges of heat flux (50, 60, 70 W/cm²) and mass flow rate (80–170 g/min) investigated in this study, CFMMC outperforms PFMMC in terms of pressure drop, temperature uniformity and comprehensive performance. Therefore, CFMMC is identified as the optimal configuration for single-phase liquid cooling systems in electronic devices operating under these conditions.

CRedit authorship contribution statement

Hao Han: Writing – review & editing, Writing – original draft, Investigation, Data curation. **Liangliang Zhang:** Writing – review & editing, Validation, Investigation. **Yuting Zhao:** Writing – review & editing, Software, Data curation. **Li Lei:** Writing – review & editing, Supervision, Project administration. **Gongming Xin:** Writing – review & editing, Supervision, Investigation, Data curation. **Jingzhi Zhang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors of the manuscript Experimental and Numerical Investigation of Single-Phase Flow and Heat Transfer Characteristics in Counter-Flow Manifold Microchannel Heat Sink submitted for publication in International Journal of Heat and Mass Transfer, disclose any actual or potential conflict of interest including any financial, personal or other relationships with other people or organizations within three years of beginning the submitted work that could inappropriately influence, or be perceived to influence, their work.

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